

Vanguard Compression Systems Ltd

ENGINEERING TECHNICAL DATASHEET & COMPLIANCE MATRIX • TENDER REF: CPCL/ENG/2026/HPGC-0412

API 618 5TH EDITION RECIPROCATING GAS COMPRESSOR DATASHEET

Design Specification: Manali Refinery High-Pressure Gas Compression System

Clause 5.4

Engineering Parameter	Tender Specified Mandatory Value	Bidder Offered Value	Compliance Status
Compressor Type & Configuration	API 618 Balanced-Opposed Reciprocating	Standard Industrial Commercial Compressor	NON-COMPLIANT
Operating Discharge Pressure	Minimum 120.0 bar(g) continuous	90.0 bar(g) Maximum Discharge (Deficit)	FAIL - DEFICIT
Suction Operating Pressure	18.5 bar(g) nominal refinery gas feed	18.5 bar(g) rating	COMPLIANT
Driver Motor Rating & Classification	Minimum 1,800 kW, Flameproof Ex d IIB+H2	1,400 kW, Non-Flameproof Enclosure	NON-COMPLIANT
Dry Gas Seal Barrier Arrangement	Dual pressurized nitrogen buffer seal	Single Mechanical Wet Seal System	NON-COMPLIANT
Vibration Monitoring System	Eddy current proximity probes per API 670	Standard Piezoelectric Accelerometers	NON-COMPLIANT

SAFETY INTEGRITY (SIL-3) & AUXILIARY SYSTEM ENGINEERING SCHEDULE

Emergency Shutdown System (ESD), Dry Gas Seals & API 614 Lubrication

Clause 5.4 Schedule B

Subsystem / Module	Engineering Design Standard	Offered Specification	Safety Compliance
Emergency Shutdown System (ESD)	IEC 61508 / 61511 SIL-3 Certification	Standard Relay Control Logic (Non-SIL)	FAILED REQUIREMENT
Lube Oil System	API 614 Package Standards	Single pump commercial lubrication pack	NON-CONFORMANCE
Pulsation Suppression Dampeners	API 618 Design Approach 3	Standard surge volume vessels	DEVIATION RECORDED
Skid Structural Engineering	Heavy duty structural steel baseplate	Lightweight channel frame	DEVIATION RECORDED

CHIEF TECHNICAL OFFICER VERIFICATION & ENGINEERING STAMP

Name: P. Ramaswamy
Designation: Chief Engineer (Turbomachinery & Process Systems)
Entity: Vanguard Compression Systems Ltd
Date & Time: 04-Sep-2026 • Verified Engineering Drawings Attached
Ref / UDIN / Reg: TECH-API618-CONF-0412

